

Two-Layer Sector Refinement of CPSV16

eq:II_ABasicTensors on the SectorBNT Sector Surface

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1 The source assertion

Cirac–Pérez–García–Schuch–Verstraete 2016 [CPGSV16] expresses any tensor A in canonical form in terms of its basis of normal tensors (BNT) A_j as

$$A^i = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1}, \quad (\text{eq:II_ABasicTensors})$$

where $\mu_{j,q} \in \mathbb{C}$ are *arbitrary* complex coefficients (no ordering or unit-modulus condition is imposed), $X_{j,q}$ are invertible gauge matrices, and A_j are the BNT normal blocks. The eventual linear independence of the BNT vectors $|V^{(N)}(A_j)\rangle$ is recorded separately in Definition `def:4:BNT` of [CPGSV21].

2 The retired specialization

An earlier formal route introduced a project-specific specialization of (`eq:II_ABasicTensors`). That route grouped equal-modulus blocks into sectors, extracted a common spectral factor, and imposed three hypotheses which are stronger than the source display:

1. **Strict spectral ordering.** The predicate requires the existence of spectral-level coefficients $\lambda_j \in \mathbb{C}$ (one per BNT sector j) with $|\lambda_0| > |\lambda_1| > \dots > |\lambda_{g-1}|$ (formerly recorded as a strict-antitone condition on the chosen spectral levels). No such ordering is present in (`eq:II_ABasicTensors`).
2. **Unit-modulus within-sector factors.** Each weight $\mu_{j,q}$ is required to factor as $\mu_{j,q} = \lambda_j \cdot \nu_{j,q}$ with $|\nu_{j,q}| = 1$, recorded by an explicit unit-modulus factorization condition. The general form (`eq:II_ABasicTensors`) allows $|\mu_{j,q}|$ to vary arbitrarily within a sector. The unit-modulus factorization is conceptually related to the RFP characterization in `thm:character-MPS` of [CPGSV16] (lines 543–555), which is a strict sub-class of the general BNT.

3. **Dominant normalization.** The dominant spectral level satisfies $|\lambda_0| = 1$. This mirrors the `IsNormalCanonicalFormBNT.mu`dom`norm`one` convention in [TNLean/MPS/BNT/Construction](#); it is not imposed by [\(eq:II_ABasicTensors\)](#).

3 The surviving formal surface

The surviving surface separates the source display into two pieces. `SectorDecomposition` stores the basis blocks, their copy multiplicities, and the raw weights $\mu_{j,q}$ through the coefficient

$$P.\text{coeff } N \ j = \sum_q (P.\text{weight } j \ q)^N.$$

`IsBNTCanonicalForm` then records the BNT hypotheses used by the fundamental-theorem route: irreducibility, left canonical normalization, gauge-phase distinctness of equal-dimensional basis blocks, eventual linear independence of the basis MPVs, bounded weights, and existence of at least one unit-modulus weight.

This keeps the two-layer information visible without assuming strict ordering of spectral levels or a factorization $\mu_{j,q} = \lambda_j \nu_{j,q}$ as part of the BNT predicate. When such a factorization is needed, it should be stated as an additional theorem hypothesis rather than hidden in the canonical-form predicate.

4 Scope restriction

The one-copy normal-canonical formulation `IsNormalCanonicalFormBNT` covers only the case $r_j = 1$ for all j . The full multi-copy BNT form with $r_j > 1$ – and the associated Newton–Girard recovery of multiplicities from $\sum_q \mu_{j,q}^N$ – is a separate mathematical gap documented in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`. The `SectorDecomposition` surface is multi-copy-ready (copies j may exceed 1), but the current adapter does not populate it beyond $r_j = 1$.

5 Formal declarations

The relevant surviving formal declarations are:

- `SectorDecomposition` - the raw two-layer sector surface ([TNLean/MPS/SharedInfra/SectorDecomposition](#)).
- `IsBNTCanonicalForm` - the BNT canonical-form predicate ([TNLean/MPS/FundamentalTheorem/SectorBNT/Basic](#)).
- `MPSTensor.IsBNTCanonicalForm.cross`overlap`basis`tendsto`zero`, `MPSTensor.IsBNTCanonicalForm.combined`family`eventually`li`, and `MPSTensor.IsBNTCanonicalForm.coeff`not`tends` in [TNLean/MPS/FundamentalTheorem/SectorBNT/Api](#).

- A future adapter from `IsNormalCanonicalFormBNT`, or from explicit separated-block hypotheses, to `IsBNTCanonicalForm`.

See also [PR #1657](#) (branch `feat/mps-ft-path-beta-pr1-two-layer`) and the audit memo `audits/2026-05-13_cpsv16_ft_path_beta_scout.md` §“PR 1.5 amendment”.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.
- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.